CUDA and OpenMP Multigrid Solvers for the Poisson Equation in 2D and 3D

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Project repository:

`https://github.com/TCC0731/multigrid-poisson-solvers`

Governing equation and discretization

Continuous model

We solve the positive-definite Poisson problem with Dirichlet boundary conditions:

$$\mathcal{L}\phi = f, \qquad \mathcal{L} = -\nabla^2,$$

on $\Omega = [0,1]^d$, with $d \in \{2,3\}$.

Discrete model

$$h = \frac{1}{n+1}, \qquad \phi_0|_{\text{interior}} = 0, \qquad \phi|_{\partial\Omega} = \phi_{\text{exact}}|_{\partial\Omega}.$$

The initial guess is zero in the interior, while boundary values are fixed by the manufactured solution. The code uses the standard 5-point stencil in 2D and 7-point stencil in 3D.

Governing equation and discretization

Dimensionless discrete operator

The interior unknowns satisfy

$$A_h\phi_h = h^2 f_h, \qquad r_h = h^2 f_h - A_h\phi_h.$$

In 2D, the standard 5-point stencil is

$$(A_h\phi)_{i,j} = 4\phi_{i,j} - \phi_{i-1,j} - \phi_{i+1,j} - \phi_{i,j-1} - \phi_{i,j+1},$$

and in 3D the 7-point stencil replaces 4 by 6 and includes the six axis-aligned neighbors.

- Dirichlet boundary data are taken from the manufactured solution; sine gives homogeneous Dirichlet data, while cosine gives inhomogeneous Dirichlet data.
- The arrays store n interior points per axis plus one boundary layer on each side, so the total size is $n + 2$.

Algorithms and backends

- Convergence criterion: the relative scaled residual

$$\eta = \frac{\|h^2 f - A_h \phi\|_2}{\|h^2 f\|_2}$$

  where A_h is the dimensionless finite-difference stencil approximating $h^2(-\nabla^2)$, must satisfy $\eta \leq \texttt{tol}$.
- Baseline smoothers: red-black SOR.
- Multigrid smoothing: red-black SOR with ν pre- and post-smoothing steps.
- If $\omega = 1$, the red-black SOR update reduces to red-black Gauss–Seidel.

Automatic SOR weight

When ω is left on auto, the code uses

$$\omega_{\text{auto}} = \frac{2}{1 + \sin\big(\pi/(n+1)\big)}.$$

Here n is the number of interior points per axis on the current grid level.

Transfer operators and cycles

- Restriction: $r_{2h} = R\, r_h$, where the 2D full-weighting stencil is

$$R_{2D} = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix},$$

  and the 3D version is the corresponding tensor-product stencil

$$(Rr)_{I,J,K} = \frac{1}{64} \sum_{a,b,c=-1}^{1} w_a w_b w_c \, r_{2I+a,\,2J+b,\,2K+c}, \qquad w_0 = 2,\ w_{\pm 1} = 1.$$

- Prolongation: $e_h = P\, e_{2h}$, where 2D uses bilinear interpolation and 3D uses trilinear interpolation; for example,

$$e_h(2I+1, 2J) = \frac{1}{2}\big(e_{2h}(I, J) + e_{2h}(I+1, J)\big), \quad e_h(2I+1, 2J+1) = \frac{1}{4} \sum_{p,q \in \{0,1\}} e_{2h}(I+p, J+q),$$

  and the 3D case uses the analogous tensor-product trilinear interpolation, involving 2, 4, or 8 coarse-grid neighbors depending on the fine-grid point location.
- Coarsest-grid solve: exact solve by default, with an RB-SOR coarse-solve variant also available.

Transfer operators and cycles

One V-cycle

- Pre-smooth on the current grid to damp high-frequency error.
- Compute the residual $r_h = h^2 f - A_h \phi$, where A_h is the dimensionless discrete Poisson stencil.
- Restrict the residual to the coarse grid: $r_{2h} = R\, r_h$.
- Solve the coarse-grid error equation recursively, $A_{2h} e_{2h} = r_{2h}$.
- Prolongate the correction and update the solution: $\phi \leftarrow \phi + P\, e_{2h}$.
- Post-smooth and repeat until the relative scaled residual satisfies $\eta \leq \texttt{tol}$.

pre-smooth $\rightarrow$ compute residual (r_h) $\rightarrow$ restrict ($r_{2h} = R r_h$)
$\rightarrow$ coarse solve ($A_{2h} e_{2h} = r_{2h}$) $\rightarrow$ prolongate/update ($\phi \leftarrow \phi + P e_{2h}$) $\rightarrow$ post-smooth

Transfer operators and cycles

W-cycle

- Start like a V-cycle: pre-smooth, form the residual, and restrict to the next coarser grid.
- Call the coarse level twice before coming back up to the fine grid.
- The repeated coarse visit reduces the remaining low-frequency error, but it also increases the per-cycle cost.

$$\text{pre-smooth} \rightarrow \text{compute residual } (r_h) \rightarrow \text{restrict } (r_{2h} = Rr_h)$$
$$\rightarrow \text{coarse cycle} \times 2 \rightarrow \text{prolongate/update } (\phi \leftarrow \phi + Pe_{2h}) \rightarrow \text{post-smooth}$$

MG-compatible grid sizes

- `--grid-size` is the number of interior points per axis, so the full array size is `grid-size + 2`.
- Nested coarsening uses $n_c = (n_f - 1)/2$, and the recursion stops when $n \leq 4$.
- Valid families: $2^a - 1$, $3 \cdot 2^a - 1$, and $5 \cdot 2^a - 1$.

Coarse-grid solve and cost

Why the bottom level matters

- The coarsest solve closes the multigrid hierarchy.
- A weak bottom-level solve can leak error back to the fine grid after prolongation.
- The choice affects both robustness and time-to-solution.

Our implementation

- Default: exact/direct solve on the coarsest grid.
- Optional: RB-SOR coarse solve for comparison or fallback.

Trade-off

- RB-SOR keeps the code path simple and is easy to benchmark.
- Exact/direct gives the strongest bottom-level correction when the coarse system is tiny.

Coarse-grid solve and cost

Coarse-grid scaling

Assume the fine-grid work is N. One level down:

$$2D : N \to N/4, \qquad 3D : N \to N/8$$

V-cycle

$$T_V \sim N + \frac{N}{2^d} + \frac{N}{2^{2d}} + \cdots$$

$$T_V^{2D} \sim \frac{4}{3}N, \qquad T_V^{3D} \sim \frac{8}{7}N$$

W-cycle

$$T_W \sim N + 2\frac{N}{2^d} + 4\frac{N}{2^{2d}} + \cdots$$

$$T_W^{2D} \sim 2N, \qquad T_W^{3D} \sim \frac{4}{3}N$$

W / V ratio and takeaway

$2D : T_W/T_V = 3/2 \approx 50\%$, $3D : T_W/T_V = 7/6 \approx 17\%$. 3D shrinks faster, so the extra W-cycle coarse-grid work matters less than in 2D.

Algorithms and backends

Implemented solvers

- Red-black SOR baseline.
- Multigrid Poisson solver in 2D and 3D.
- Both V-cycle and W-cycle variants.
- Coarsest-grid solve with SOR or exact/direct solve.

Implemented support

- Python backend.
- OpenMP CPU backend.
- CUDA GPU backend.
- Full-weighting restriction and bilinear/trilinear prolongation.
- Manufactured-solution Dirichlet boundary setup.

Verified cases

Case	2D exact solution	3D exact solution
`sine`	$\sin(\pi x)\sin(\pi y)$	$\sin(\pi x)\sin(\pi y)\sin(\pi z)$
`mixed_sine`	$\sin(2\pi x)\sin(3\pi y)$	$\sin(2\pi x)\sin(3\pi y)\sin(4\pi z)$

Implementation optimizations

OMP

- 'parallel for' + 'simd' on the hot stencil loops.
- Focus on CPU core utilization and vectorization.
- Goal: make the smoother and transfer operators scale better on the host.

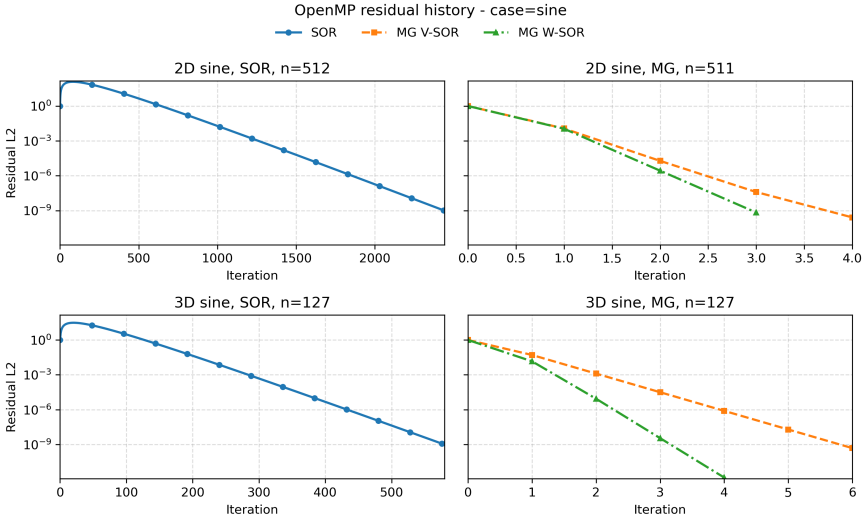
CUDA

- Fuse residual and restriction to cut global memory traffic.
- Pool MG workspace and capture the iteration loop with CUDA Graph.
- Use fused small-grid kernels on coarse levels to avoid repeated launches.

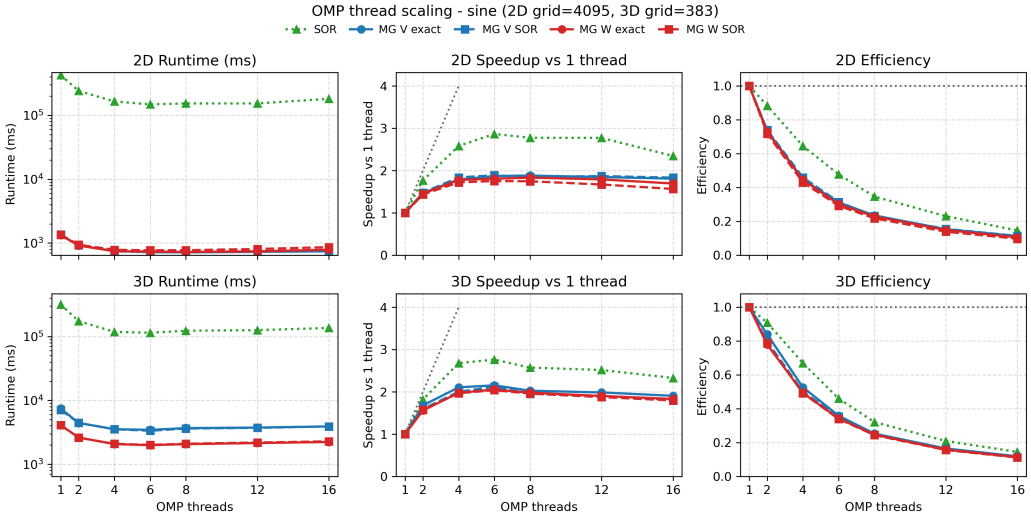
Note

Red-black ordering is an enabler for parallel updates, not the main optimization itself.

Empirical results



Empirical results



Bottleneck analysis

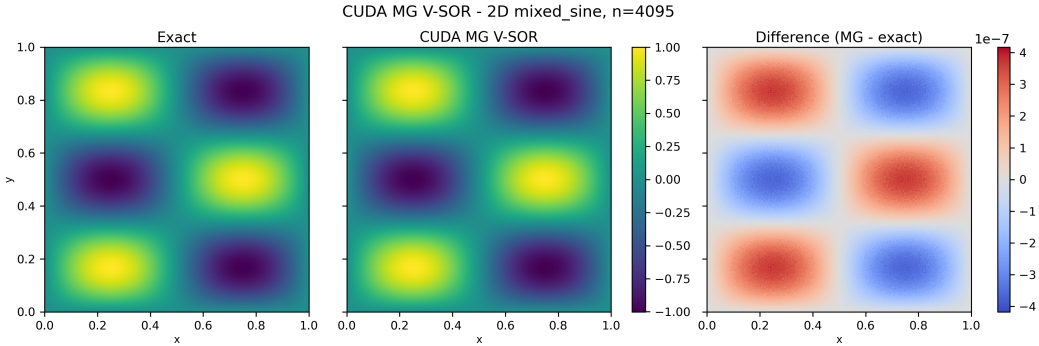
Topdown, L1, and cache evidence

- The OpenMP solver is backend-bound, not compute-bound.
- `topdown-be-bound` is 57.7% in 2D and 58.2% in 3D.
- `topdown-mem-bound` rises from 43.7% in 2D to 49.3% in 3D.
- L1D miss rate more than doubles, from 1.5% in 2D to 3.2% in 3D.
- The 3D LLC load miss rate is 89.4%, much higher than the 2D value of 26.4%.

Profile takeaway

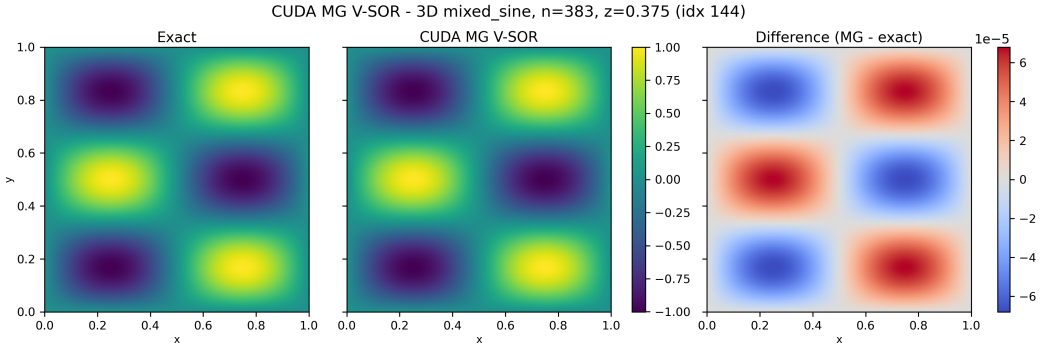
- The hot kernel is `smooth_red_black` / `smooth_red_black_3d`.
- `perf annotate` shows the inner loop dominated by grid-value loads and `rhs` fetches.
- The kernel has low arithmetic intensity and spends most of its time waiting on memory.
- Conclusion: poor OpenMP thread scaling is mainly caused by memory bandwidth and cache pressure.

Mixed-sine test cases



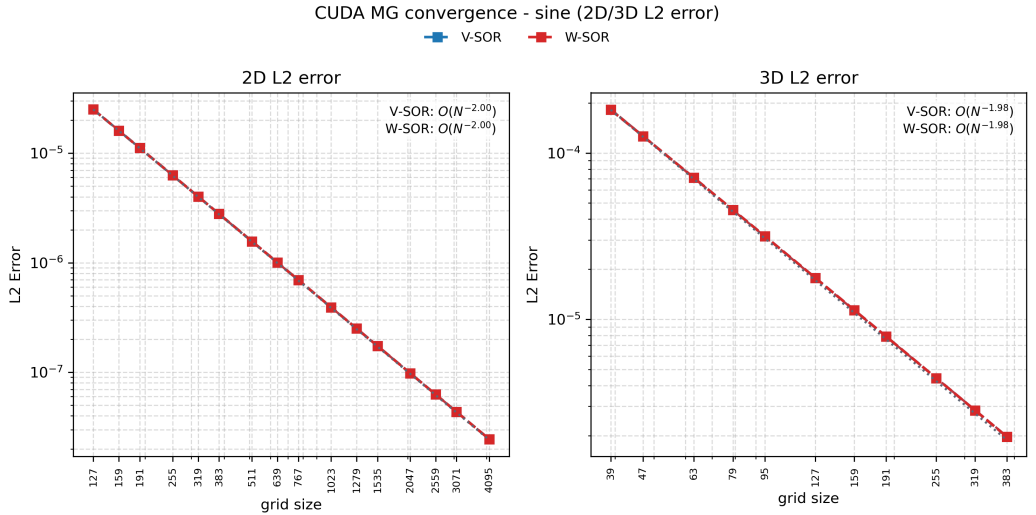
CUDA MG V-SOR on the manufactured solution $\phi_{\text{exact}}(x, y) = \sin(2\pi x)\sin(3\pi y)$ with $\nu = 3$ and $\omega = 1.25$.

Mixed-sine test cases



CUDA MG V-SOR on the manufactured solution $\phi_{\text{exact}}(x, y, z) = \sin(2\pi x)\sin(3\pi y)\sin(4\pi z)$ with $\nu = 3$ and $\omega = 1.25$.

Convergence analysis



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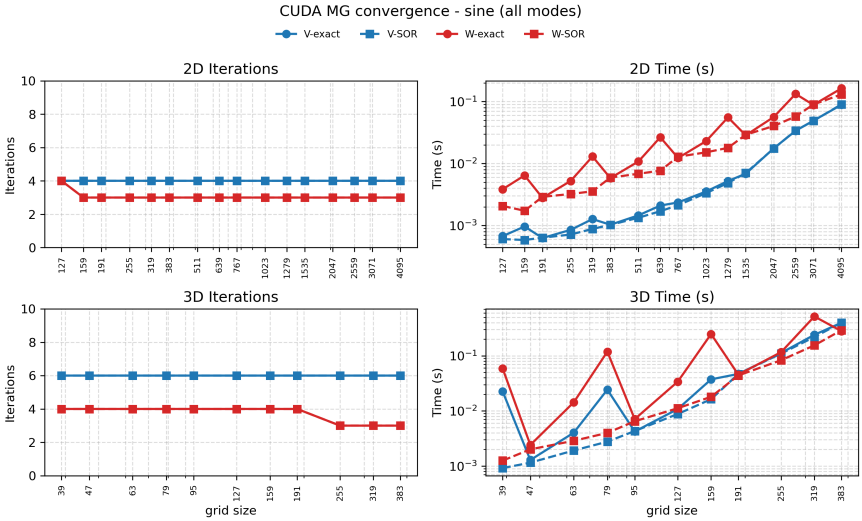
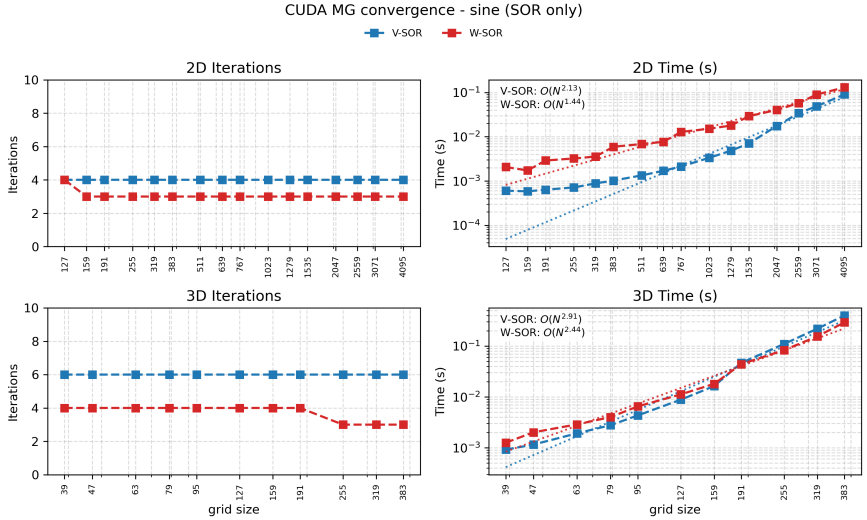
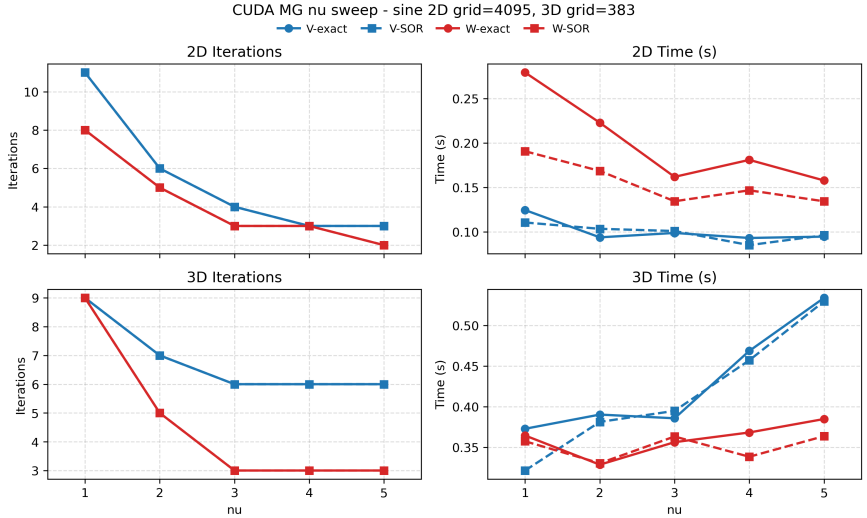


Convergence analysis



Parameter sweeps



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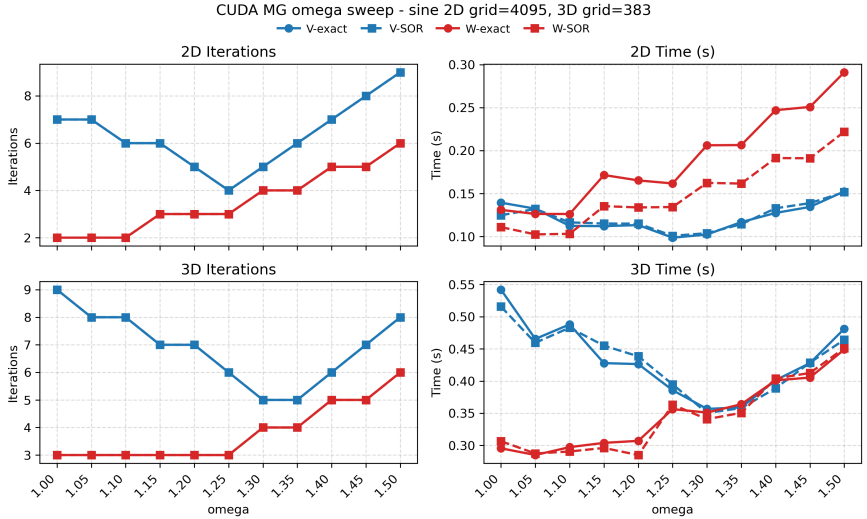

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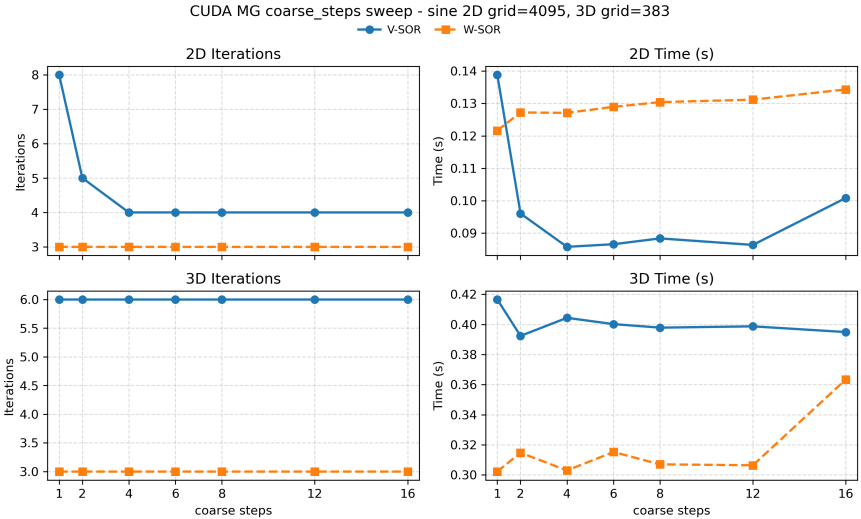
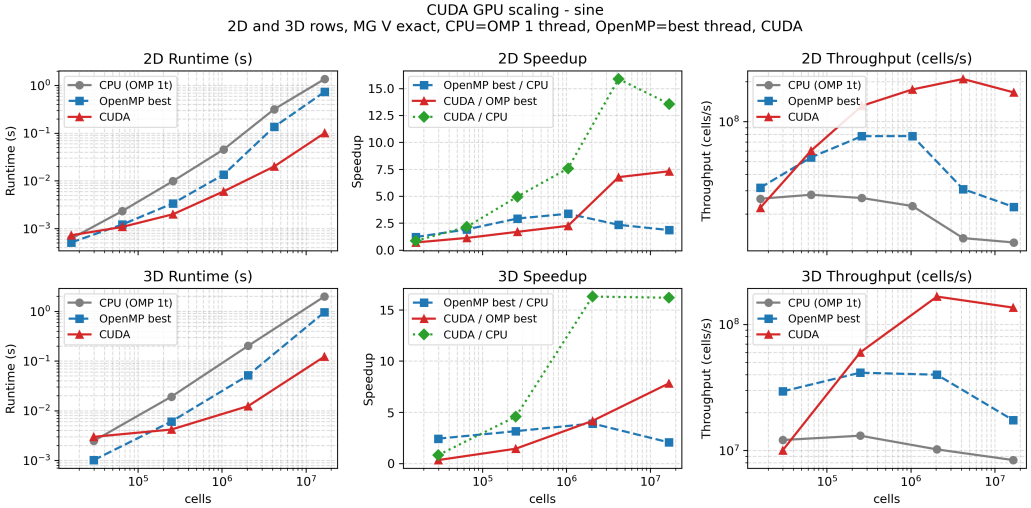
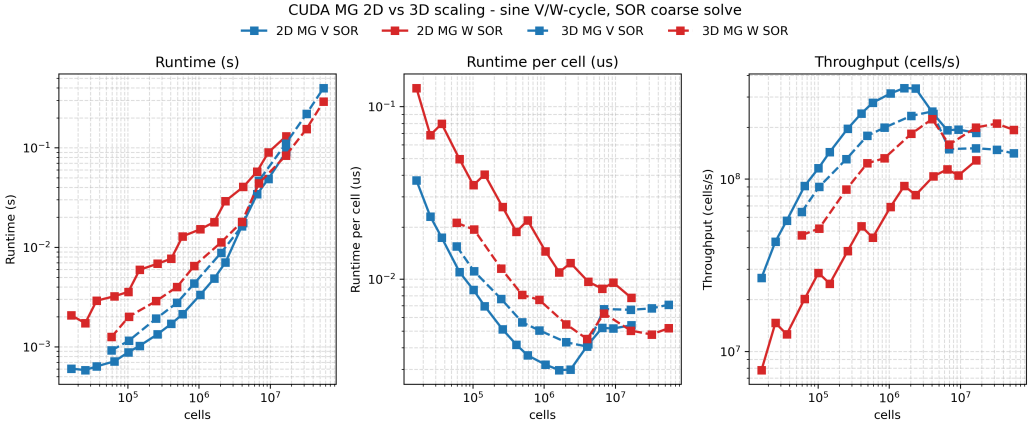


Performance scaling



Performance scaling



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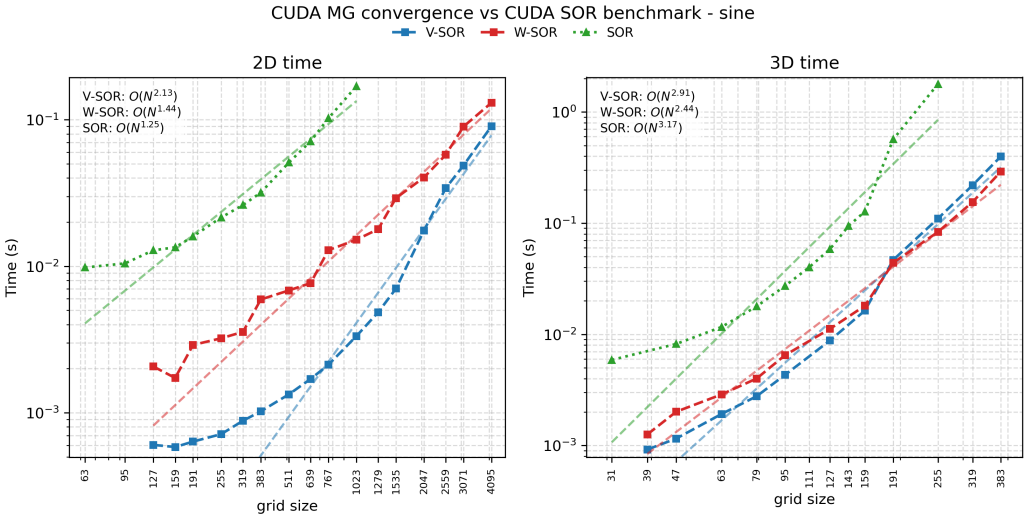

Conclusions

What we learned

- The CUDA multigrid solver converges reliably in both 2D and 3D on manufactured-solution tests.
- The ν and ω sweeps show that solver time is sensitive to smoother tuning, especially in 2D.
- Compared with SOR, multigrid gives the better overall time-to-solution story as the grid grows.
- The OpenMP backend flattens because the hot smoothing kernel is memory-bound rather than compute-bound.

Bottom line

Multigrid is the right algorithmic direction here, CUDA gives the strongest performance path, and the remaining CPU scaling limit is mainly a memory-bandwidth problem.

Conclusions

- Numerical Recipes, 3rd ed., Section 20.6, *Multigrid Methods for Boundary Value Problems*.
- Achi Brandt and Oren E. Livne, *Multigrid Techniques: 1984 Guide with Applications to Fluid Dynamics, Revised Edition* (SIAM, 2011).
  `https://epubs.siam.org/doi/book/10.1137/1.9781611970753`

Q&A

Thank you